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        "# Risk Heat Map - Apple Inc Acquiring Netflix\n",
        "## ALY 6130 | Group Members: Samuel Tweneboah-Kodua Nyamekye |  

        Adebukola Fadina | Sandeep Kaur"
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        "import matplotlib.pyplot as plt\n",
        "import matplotlib.patches as mpatches\n",
        "import numpy as np"
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        "### Step 2 - Enter the risk data from the risk calculation sheet\n",
        "These scores come directly from the risk calculation sheet.  

        Likelihood and Impact use the professor's scale (1, 3, 5, 7, 9 for  

        likelihood and 1, 2, 4, 6, 8, 9 for impact)."
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        "# Risk data - taken from the risk calculation sheet\n",
        "# Format: (Risk ID, Risk Name, Likelihood Score, Impact Score)\n",
        "\n",
        "risks = [\n",
        "    ('R-001', 'Government\\nBlocking Deal',          7, 9),\n",
        "    ('R-002', 'Studios Pull\\nContent',              7, 8),\n",
        "    ('R-003', 'Growing Faster\\nTogether',           7, 8),\n",
        "    ('R-004', 'Netflix Data\\nLawsuit',              7, 8),\n",
        "    ('R-005', 'Losing Key\\nPeople',                 7, 6),\n",
        "    ('R-006', 'Moving Off\\nAWS',                    5, 8),\n",
        "    ('R-007', 'Global Growth\\nChance',              7, 8),\n",
        "    ('R-008', 'People Cancelling\\nOver Prices',     7, 6),\n",
        "]\n",
        "\n",
        "# Calculate risk scores (Likelihood x Impact)\n",
        "for r in risks:\n",
        "    score = r[2] * r[3]\n",
        "    print(f"{r[0]} - {r[1].replace(chr(10), ' ')} | Likelihood={r[2]} | Impact={r[3]} | Score={score}")
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        "### Step 3 - Set the thresholds\n",
        "These thresholds come from the risk calculation sheet scores, not a generic 1 to 10 scale."
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        "# Thresholds from the risk calculation sheet\n",
        "LOW_MAX = 20    # Score 1-20 = Low Risk (Green)\n",
        "MEDIUM_MAX = 54 # Score 21-54 = Medium Risk (Yellow)\n",
        "HIGH_MIN = 55   # Score 55-81 = High Risk (Red)\n",
        "\n",
        "print('Thresholds set:')\n",
        "print(f'Low Risk = Score 1 to {LOW_MAX} --> Green')\n",
        "print(f'Medium Risk = Score 21 to {MEDIUM_MAX} --> Yellow')
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    "print(f'High Risk    = Score {HIGH_MIN} to 81    --> Red')"
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        "# Likelihood values on X axis (professor's scale)\n",
        "likelihood_vals = [1, 3, 5, 7, 9]\n",
        "\n",
        "# Impact values on Y axis (professor's scale)\n",
        "impact_vals = [1, 2, 4, 6, 8, 9]\n",
        "\n",
        "fig, ax = plt.subplots(figsize=(14, 9))\n",
        "\n",
        "# Draw the background color zones\n",
        "for i, imp in enumerate(impact_vals):\n",
        "    for j, lik in enumerate(likelihood_vals):\n",
        "        score = lik * imp\n",
        "        if score >= HIGH_MIN:\n",
        "            color = '#FF4444'    # Red = High\n",
        "        elif score >= LOW_MAX + 1:\n",
        "            color = '#FFD700'    # Yellow = Medium\n",
        "        else:\n",
        "            color = '#70AD47'    # Green = Low\n",
        "\n",
        "        rect = mpatches.FancyBboxPatch(\n",
        "            (j - 0.45, i - 0.45), 0.9, 0.9,\n",
        "            boxstyle='round,pad=0.02',\n",
        "            facecolor=color, edgecolor='white', linewidth=1.5\n",
        "        )\n",
        "        ax.add_patch(rect)\n",
        "\n",
        "        # Show the score inside each cell\n",
        "        ax.text(j, i, str(score),\n",
        "            ha='center', va='center',\n",
        "            fontsize=8, color='white', fontweight='bold')\n",
        "\n",
        "# Plot the risks as dots with labels\n",
        "for risk_id, name, lik, imp in risks:\n",
        "    score = lik * imp\n",
        "\n",
        "    # Find position on grid
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"    x = likelihood_vals.index(lik)\n",
"    y = impact_vals.index(imp)\n",
"\n",
"    # Dot color based on score\n",
"    if score >= HIGH_MIN:\n",
"        dot_color = '#8B0000'\n",
"    elif score >= LOW_MAX + 1:\n",
"        dot_color = '#B8860B'\n",
"    else:\n",
"        dot_color = '#006400'\n",
"\n",
"    ax.scatter(x, y, s=200, color=dot_color,\n",
"               zorder=5, edgecolors='black', linewidth=1)\n",
"\n",
"    ax.annotate(f"\{risk_id}\n\{name}",\n",
"               xy=(x, y),\n",
"               xytext=(x + 0.15, y + 0.15),\n",
"               fontsize=6.5,\n",
"               color='black',\n",
"               fontweight='bold',\n",
"               bbox=dict(boxstyle='round,pad=0.2',\n",
"                           facecolor='white',\n",
"                           edgecolor='gray',\n",
"                           alpha=0.8))\n",
"\n",
"# Set axis labels and ticks\n",
"ax.set_xticks(range(len(likelihood_vals)))\n",
"ax.set_xticklabels([str(v) for v in likelihood_vals],\n",
"                   fontsize=11)\n",
"ax.set_yticks(range(len(impact_vals)))\n",
"ax.set_yticklabels([str(v) for v in impact_vals], fontsize=11)\n",
"\n",
"ax.set_xlabel('Likelihood Score', fontsize=13, fontweight='bold',\n",
"              labelpad=10)\n",
"ax.set_ylabel('Impact Score', fontsize=13, fontweight='bold',\n",
"              labelpad=10)\n",
"\n",
"ax.set_title('Risk Heat Map - Apple Inc Acquiring Netflix\n\ALY 6130\n",
"| Samuel Tweneboah-Kodua Nyamekye | Adebukola Fadina | Sandeep Kaur',\n",
"            fontsize=13, fontweight='bold', pad=15)\n",
"\n",
"# Legend\n",
"high_patch = mpatches.Patch(color='#FF4444', label='High Risk\n",
"(Score 55-81)')\n",
"medium_patch = mpatches.Patch(color='#FFD700', label='Medium Risk\n",
"(Score 21-54)')\n",
"low_patch = mpatches.Patch(color='#70AD47', label='Low Risk\n",
"(Score 1-20)')\n",
"ax.legend(handles=[high_patch, medium_patch, low_patch],\n",
"          loc='lower right', fontsize=10, framealpha=0.9)\n",
"\n",
"ax.set_xlim(-0.5, len(likelihood_vals) - 0.5)\n",
"ax.set_ylim(-0.5, len(impact_vals) - 0.5)\n",
"ax.grid(False)

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        "\n",
        "plt.tight_layout()\n",
        "plt.savefig('risk_heatmap.png', dpi=150, bbox_inches='tight')\n",
        "plt.show()\n",
        "print('Heat map saved as risk_heatmap.png')"
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 "### Step 5 - Print the full risk summary table"

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 "print('='*70)\n",

 "print('RISK SCORE SUMMARY - Apple Inc Acquiring Netflix')\n",

 "print('='*70)\n",

 "print(f\"{'Risk ID':<8} {'Risk Name':<35} {'L':>3} {'I':>3}

{'Score':>6} {'Priority':<10} {'Zone'}\")\n",

 "print('-'*70)\n",

 "\n",

 "for risk_id, name, lik, imp in risks:\n",

 " score = lik * imp\n",

 " if score >= HIGH_MIN:\n",

 " priority = 'H'\n",

 " zone = 'HIGH'\n",

 " elif score >= LOW_MAX + 1:\n",

 " priority = 'M'\n",

 " zone = 'MEDIUM'\n",

 " else:\n",

 " priority = 'L'\n",

 " zone = 'LOW'\n",

 "\n",

 " clean_name = name.replace('\n', ' ')\n",

 " print(f\"{risk_id:<8} {clean_name:<35} {lik:>3} {imp:>3}

{score:>6} {priority:<10} {zone}\")\n",

 "\n",

 "print('='*70)\n",

 "print(f'Thresholds: Low = 1-{LOW_MAX} | Medium = 21-{MEDIUM_MAX} |

High = {HIGH_MIN}-81')"

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